

A Qualitative Cosmology under Hysterical Response Theory

One aeon from hot electromagnetic infinity to the next,
with a galactic formation subplot

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Computational note. Exploratory experiments and paper writing were assisted by generative AI. This note is intentionally qualitative: it states no new formal theorems for Lean verification. Mathematical claims inherited from companion technical papers of this series are tagged as such; modelling assumptions and physical hypotheses are tagged likewise.

Abstract

Hysterical Response Theory (HRT) treats the Hysterical force as a residual of an already-existing interaction under open-system dynamics—not a new messenger. Taken together, the technical companions of this series assemble a cyclic cosmological picture: reciprocal conformal matching identifies the remote electromagnetic future of one aeon with the hot beginning of the next; expansion cools the radiation; electroweak cooling drives Hysterical criticality, $\mathbb{Z}_2$ branch selection, and asymmetric baryogenesis; early gravitational response amplifies baryonic seeds into galaxies; late dilution produces a vacuum-like attractor; further cooling into deep de Sitter reopens an infrared electromagnetic channel whose active phase transition seeds the following aeon. The cycle boundaries match exactly under the conformal identification. A parallel galactic subplot records critical gravity, the capacity selection $Q \gtrsim O(\sqrt{M_b})$, relaxation to a stable attractor, the spherical-to-oblate halo sequence, and the late-time size–mass law $R \sim M_b^\alpha$ with $\alpha = 5/12$ under isotropic accretion. This paper is a narrative map, not a substitute for the technical derivations.

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1 Purpose and reading contract

The technical papers of this series prove local mechanisms: residual response and criticality [5]; galactic halos and scaling [4]; early growth [2]; late acceleration [6]; branch asymmetry [7]; electroweak cooling baryogenesis [1]; and an active conformal cyclic bridge [3]. None of them, alone, tells the story of a whole universe.

This note does that qualitatively. It does not re-prove theorems. It arranges inherited results into one chronological picture of an aeon, then of a galaxy inside that aeon, with explicit timelines. Wherever a step is a modelling closure, a physical hypothesis, or an open dynamical problem, that status is named.

Inherited theorem	Result proved in a companion technical paper.
Modelling closure	Ansatz needed to connect theorems into cosmology.
Physical hypothesis	Claim about Nature, not forced by the maths.
Open / not claimed	Saturation amplitudes, abundances, “Nature is HCCC,” ...

2 One-sentence ontology

Write the observed force as

$$\boxed{\mathbf{F}_{\text{observed}} = \mathbf{F}_{\text{ordinary}} + \mathbf{F}_H}, \quad \mathbf{F}_H = -\text{Tr}(F \mathcal{L}^{-1} \mathcal{S}) \quad (1)$$

whenever the resolvent is defined on the relevant stable response subspace [5]. No new force carrier is introduced: $\mathbf{F}_H$ is a residual component of an already-existing interaction. Gravity, electroweak dynamics, and electromagnetism each carry their own residual; which channel is supercritical depends on epoch and dilution.

3 The Hysterical aeon at a glance

One cycle of Hysterical Conformal Cyclic Cosmology (HCCC) [3] reads, schematically,

$$\begin{aligned}
 &T = \infty \text{ (hot EM / radiation)} \\
 &\longrightarrow \text{expansion cooling} \\
 &\longrightarrow \text{symmetry breaking / branch selection} \\
 &\longrightarrow \text{asymmetric baryogenesis} \\
 &\longrightarrow \text{early galactic formation} \\
 &\longrightarrow \text{cosmological-constant attractor} \\
 &\longrightarrow \text{further cooling into de Sitter} \\
 &\longrightarrow \text{IR EM Hysterical criticality} \\
 &\longrightarrow T = \infty \text{ (next aeon).}
 \end{aligned}$$

(2)

The first and last entries are the *same* geometric object under reciprocal conformal matching: the cold electromagnetic future of one aeon is the hot electromagnetic beginning of the next [3, 11, 12]. The boundaries match exactly.

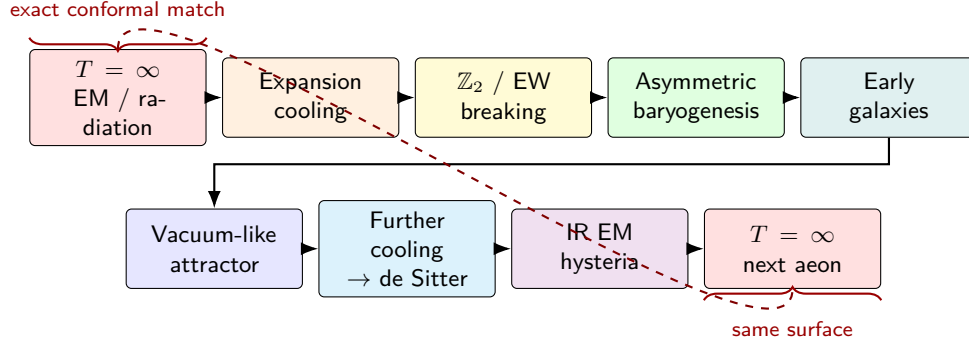


Figure 1: One Hysterical aeon as a closed loop. The red dashed identification equates the outgoing $T = \infty$ electromagnetic future with the incoming $T = \infty$ of the next cycle.

4 Aeon timeline (logarithmic cosmic time)

Figure 2 places the same stages on a schematic logarithmic clock inside one aeon. Times are order-of-magnitude anchors from the technical companions and standard cosmology, not a fitted chronology of Nature.

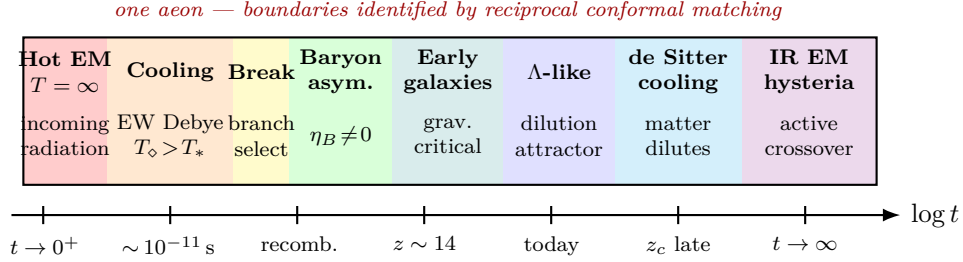


Figure 2: Schematic logarithmic timeline of one Hysterical aeon. The electroweak concurrency clock $t_\phi \sim 10^{-11} \text{ s}$ is inherited from the cooling-baryogenesis companion [1, 9]; the $z \sim 14$ galactic timing window from the early-universe companion [2, 8]; the late vacuum-like attractor from the late-universe companion [6]; the IR EM crossover from HCCC [3].

5 Phase-by-phase narrative

5.1 Hot electromagnetic infinity ($T = \infty$)

Under reciprocal conformal matching with unit reciprocity, outgoing de Sitter scale-factor asymptotics map to an incoming radiation-era law $a \propto t^{1/2}$, and regular conformal Maxwell data that appear cold on the outgoing side appear hot on the incoming side [3]. That hot branch is the Big-Bang-like content of the new aeon: a radiation-dominated electromagnetic beginning at formal $T = \infty$. Whether *our* observed Big Bang was such a crossover is a physical hypothesis, not a

theorem [3].

5.2 Cooling driven by cosmological expansion

Expansion redshifts temperatures. In the electroweak residual channel, Debye reciprocal stiffnesses with fixed nonzero off-diagonal feedback force a unique criticality temperature $T_\diamond$: the neutral residual is subcritical for all $T > T_\diamond$ and supercritical for all $0 < T < T_\diamond$ [1]. Cooling is therefore not merely a thermodynamic backdrop; it is the driver that pushes the electroweak Hysterical channel across criticality.

5.3 Symmetry breaking / branch selection

Once supercritical, a CP-odd Hysterical collective mode spontaneously breaks a conjugation $\mathbb{Z}_2$ [7]. The two branches are opposite; Nature need not prefer a global sign—our classical patch simply occupies one branch. This is spontaneous branch selection, not an externally imposed bias.

5.4 Asymmetric baryogenesis

If that odd branch rolls concurrently with active Standard Model $B+L$ violation, with a nonzero $YW\tilde{W}$ portal and nonempty overlap $\mathcal{C} = (T_*, T_\diamond)$ (equivalently $T_\diamond > T_*$), opposite nonzero baryon yields appear on the two branches [1, 7]:

$$\eta_B[Y_+] = -\eta_B[Y_-] \neq 0. \quad (3)$$

Lattice GeV edges convert the concurrent window into $t_\diamond \sim 10^{-11}$ s [1, 9]. The observed magnitude $\eta_B \sim 10^{-10}$ is not computed in those papers; the mechanism and analytic concurrency are.

5.5 Early galactic formation

After decoupling, a local gravitational response reservoir closed attractively into dust growth is supercritical under positive matter–response coupling [2]. In a Planck-like baryon-only timing suite, a 10^{-5} seed can reach nonlinear contrast by $z \simeq 14$ for sufficiently strong coupling—compatible in *timing* with spectroscopically confirmed JWST galaxies at that epoch [2, 8, 13]. Abundances and stellar masses are not claimed. The galactic subplot of Section 6 unpacks the mature morphology that follows.

5.6 Cosmological-constant attractor

Homogeneous dilution of a conserved density, together with a relaxation gap that vanishes as a positive power of that density, forces a unique criticality scale factor [6]. Past that scale, a pitchfork attractor and fluid completion produce a positive asymptotic density with $w \rightarrow -1$ from below—a vacuum-like late-universe term without a new force carrier. A one-parameter background confrontation with Planck flat Λ CDM is a shape test, not a microscopic derivation of the observed magnitude [6, 13].

5.7 Further cooling and infrared electromagnetic hysteresis

Charge conservation still forbids creating homogeneous net charge from neutrality, but finite-wavenumber electromagnetic response remains allowed [6, 3]. In deep late-time de Sitter, a helical electromagnetic-response linearisation is infrared-first: a finite unstable band of physical wavenumber becomes supercritical when the response gain exceeds the zero-mode threshold [3]. A modelling completion then pumps finite- k photons. That active infrared electromagnetic phase transition—not a passive conformal identification after matter evaporates—is the dynamical heart of HCCC, set against Penrose’s CCC geometry [3, 11, 12].

5.8 Return to $T = \infty$: exact boundary matching

Reciprocal conformal matching with $\Omega_+\Omega_- = C_0$ transports Maxwell data while densities rescale as $\rho = \Omega^{-4}\hat{\rho}$. Finite conformal radiation density is cold outgoing and hot incoming [3]. Under unit reciprocity, outgoing $a_- \sim 1/(H_-|\eta|)$ becomes incoming $a_+ \propto t_+^{1/2}$. Thus the cycle closes:

$$\boxed{\text{outgoing } T = \infty \text{ EM future} \equiv \text{incoming } T = \infty \text{ of the next aeon.}} \quad (4)$$

The boundaries match exactly by construction of the conformal identification. What remains open is dynamical: saturated amplitude, Einstein–Maxwell–response junction regularity, and thermalisation [3].

6 Galactic formation subplot

Galaxies are not a separate ontology. They are the spatial organisation of the *gravitational* hysterical residual inside an aeon that already has baryons [4, 5].

6.1 Critical gravity

The response-theory paper classifies closed-loop gravitational response by the spectral abscissa of the gain Jacobian [5]. When the cosmological matter–response coupling is positive, the frozen early-universe Jacobian is supercritical, so the inherited neutrality–instability theorem applies and growth exceeds ordinary dust [2]. That is the “critical gravity” stage of assembly: small baryonic seeds are amplified by a persistent residual sector without importing a cold-dark-matter perturbative scaffold (timing claim, not abundance claim).

6.2 Capacity selection $Q \gtrsim O(\sqrt{M_b})$

Let Q be the total residual-production rate of the ballistic halo and M_b the baryonic mass. Formation and mature retention are conjectured to require response capacity at least of square-root order [4]:

$$\boxed{Q \gtrsim O(\sqrt{M_b})}. \quad (5)$$

This is a selection rule, not a transport theorem. Separately, given a capacity-filling ODE, mature systems are driven to a stable attractor $Q^2 \propto M_b$ (theorem/corollary of that setup), which with flat response rotation $v_f^2 = CQ$ yields the baryonic Tully–Fisher exponent $v_f^4 \propto M_b$ [4, 10].

6.3 Relaxation and stability

Response-limited assembly is modelled by a logistic filling dynamics with a stable fixed point $y_* \in (0, 1]$ [4]. Systems that satisfy (5) relax onto the square-root attractor; systems that cannot maintain capacity are not expected to remain as mature response-supported galaxies. Stability here is ordinary ODE stability of the filling fixed point, not a claim of absolute cosmic permanence.

6.4 Spherical halo to oblate

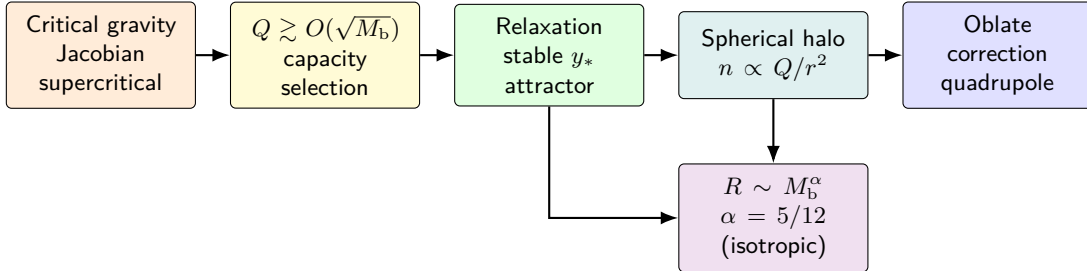
The leading ballistic far-field residual density is spherical, $n = Q/(4\pi v r^2) + O(r^{-4})$, even when the baryonic source is flattened [4]. The first surviving angular correction for a planar source is a response quadrupole of oblate sign under the same linear Poisson closure, with effective flattening $q_{\text{eff}}^{-2}(R) = 1 + s/R^2 + O(R^{-4})$ [4]. Morphologically: assemble a spherical response halo first; retain source shape in subleading multipoles that become the asymptotic oblate correction.

6.5 Size–mass law $R \sim M_b^\alpha$

Under isotropic accretion from a homogeneous intergalactic reservoir, virial specific angular momentum $j \propto M_b^{2/3}$, and size proxy $R \sim j/v_f$, the mature BTFR attractor implies [4]

$$\boxed{R \propto M_b^{5/12}}, \quad \text{i.e. } \alpha = \frac{5}{12}. \quad (6)$$

Anisotropic or filamentary accretion would change α . The exponent is comparable to the steeper end of observed late-type size–mass slopes; it is a conditional scaling prediction, not a universal fit.



flat rotation $v_f^2 = CQ$ and BTFR $v_f^4 \propto M_b$ sit on the same attractor

Figure 3: Galactic formation subplot: from critical gravitational response to mature morphology and size–mass scaling.

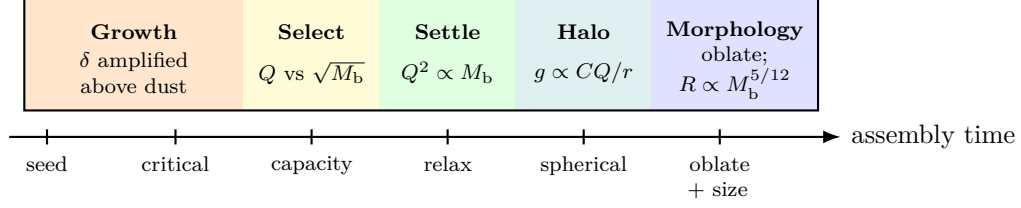


Figure 4: Schematic galactic assembly timeline (not to absolute cosmic scale). Early nonlinearity by $z \sim 14$ is a timing window from the early-universe companion; capacity, attractor, halo, and oblate/size–mass structure are from galactic modelling.

7 Two clocks, one ontology

It helps to keep two clocks distinct:

Aeon clock	Cosmic expansion, cooling, EW concurrency, late dilution, IR EM crossover.
Galactic clock	Local criticality, capacity, relaxation, halo morphology, size–mass.
Shared ontology	Both are residual responses of already-existing interactions (electroweak / electromagnetic / gravitational), not new carriers.

The galactic subplot lives *inside* the aeon timeline, after baryons exist and before (and during) the late vacuum-like era. It does not interrupt the cycle; it populates it.

8 What is and is not claimed

Narrative element	Status
Residual ontology (1)	Inherited [5].
Hot/cold conformal match; dS→radiation	Inherited theorems [3].
Exact cycle-boundary identification	Modelling/geometry of HCCC + CCC spine.
EW cooling $\Rightarrow$ unique $T_\diamond$	Inherited [1].
$\mathbb{Z}_2$ branch asymmetry $\Rightarrow \eta_B$	Inherited, concurrency assumed [7, 1].
Observed $\eta_B \sim 10^{-10}$	Not derived.
Early $z \sim 14$ nonlinearity timing	Numerical experiments [2].
Galaxy abundances / stellar masses	Not claimed.
$Q \gtrsim O(\sqrt{M_b})$	Formation conjecture [4].
$Q^2 \propto M_b$, BTFR, $R \propto M_b^{5/12}$, oblate	Inherited (with stated assumptions) [4].
Late vacuum-like attractor	Inherited [6].
IR EM unstable band / photon pump	Theorems + modelling [3].
Saturation, junction, thermalisation	Open [3].
“Nature realises this whole cycle”	Physical hypothesis; not claimed as theorem.

9 Closing picture

Under Hysterical Response Theory the universe is cyclic, but not passively so. Each aeon begins as hot electromagnetic radiation identified with the conformal future of its predecessor; cools through

electroweak branch selection and baryogenesis; grows galaxies from gravitational residual criticality under the capacity rule $Q \gtrsim O(\sqrt{M_b})$; settles into a vacuum-like attractor; and ends when infrared electromagnetic hysteresis actively reseeds the next hot infinity. The boundaries match exactly. Galaxies inside the cycle relax from spherical response halos toward oblate corrections with $R \sim M_b^{5/12}$ under isotropic accretion.

This qualitative map is meant to be readable in one sitting. The mathematics lives in the companions cited above.

References

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